

# Time-Aware Transformer for Cross-Hospital Sepsis Early Warning: Linking Discrimination, Decision Utility, and Alert Burden

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September 5, 2026

## Abstract

**Background:** Conventional predictive metrics such as AUROC and AUPRC are standard benchmarks for evaluating clinical machine learning models. However, in temporal early-warning applications—such as forecasting sepsis onset in intensive care units (ICUs)—high discriminative rank-ordering does not automatically guarantee optimal net clinical utility when decision thresholds and false alarm burdens are considered under operational deployment.

**Objective:** We systematically develop and evaluate a progression of deep learning architectures for early sepsis prediction across hospital systems, establish the strongest discriminative representation, and evaluate net clinical utility under the official PhysioNet 2019 metric while auditing operational alert workload.

**Methods:** We evaluated a structured family of predictive models—ranging from classical gradient boosting (*M1*) and plain Transformers (*M2*) to Time-Aware Transformers (*M3*) incorporating physiological values, missingness masks, and elapsed-time deltas ( $\Delta t$ ), alongside organ-aware (*M4*) and multi-hybrid routing architectures (*M5*). Models were trained on 20,336 ICU stays from Beth Israel Deaconess Medical Center (Set A / BIDMC; partitioned into 18,302 training stays [90.0%] and 2,034 validation stays [10.0%]) and evaluated on an independent held-out test cohort of 20,000 ICU stays from Emory University Hospital (Set B / Emory; 1,066 septic patients [5.33%], 18,934 non-septic patients [94.67%], 753,927 hourly observations). Controlled factorial ablation experiments were conducted to isolate missingness and temporal encodings. Decision threshold selection was strictly isolated: candidate thresholds were evaluated on the BIDMC validation split ( $N = 2,034$ ), prespecifying  $th = 0.190$  prior to unblinding external test predictions. Net clinical utility was evaluated using the official PhysioNet 2019 evaluation script (`evaluate_sepsis_score.py`), incorporating a 6-hour onset lead time shift ( $t_{\text{sepsis}} = t_{\text{label}} + 6\text{h}$ ) and metric normalization  $U_{\text{official}} = (U_{\text{obs}} - U_{\text{inact}}) / (U_{\text{best}} - U_{\text{inact}})$ . Uncertainty was quantified via patient-level cluster bootstrap resampling ( $B = 1,000$ ) and multi-seed testing ( $N = 6$  seeds).

**Results:** Among the evaluated architectures, the compact *M3* Time-Aware Transformer provided the strongest cross-hospital discriminative representation (AUROC = 0.961726 [0.9617], AUPRC = 0.423114 [0.4231], Brier = 0.015290, ECE = 0.018151), outperforming XGBoost (*M1*, AUROC = 0.8842), plain Transformers (*M2*, AUROC = 0.9265), organ-aware (*M4*, AUROC = 0.9582), and multi-hybrid models (*M5*, AUROC = 0.9591). Factorial ablations demonstrated significant main effects for missingness masks (+0.0155 AUROC) and time deltas (+0.0215 AUROC), while increasing architectural complexity beyond *M3* (*M4*, *M5*) did not provide statistically significant improvement ( $p = 0.068$ ). Evaluated under the official PhysioNet 2019 Challenge metric on Emory external test data, the decision threshold  $th = 0.190$ —prespecified strictly on BIDMC validation data before external test evaluation—achieved an Official Normalized Utility of  $U_{\text{official}} = +0.655944$  (+0.6559, 95%

CI:  $[+0.6310, +0.6800]$ ), coinciding with the highest observed utility in post-hoc descriptive Emory sensitivity analysis. Operational workload auditing revealed an alert frequency of 16.99 alerts per 100 patient-days (5,337 total alerts: 1,004 true positive, 4,333 false positive; alert Positive Predictive Value [PPV] = 18.81%; 25.86% of patients alerted).

**Conclusion:** The Time-Aware Transformer (*M3*) demonstrated superior cross-hospital discrimination (0.9617) and positive net utility (+0.6559) under official PhysioNet evaluation, while operational workload auditing revealed a substantial false alert burden (81.19% false alarms) under cross-hospital prevalence shift. Positive utility under the PhysioNet 2019 scoring framework reflects net benefit under a predefined challenge cost function but does not constitute direct proof of prospective clinical effectiveness. These findings emphasize that decision-theoretic evaluation, workload auditing, and threshold isolation are essential prior to prospective clinical deployment.

## Introduction

Early recognition of sepsis in intensive care units (ICUs) is a major imperative in acute care medicine. Sepsis, defined as life-threatening organ dysfunction caused by a dysregulated host response to infection (Singer et al., 2016), affects over 49 million individuals annually and accounts for nearly 20% of global hospital deaths.

In clinical machine learning literature, predictive performance is overwhelmingly benchmarked using conventional rank-ordering metrics, primarily AUROC and AUPRC. While these metrics measure an algorithm’s capacity to rank high-risk observations above low-risk observations across arbitrary decision thresholds, they ignore operational deployment realities—such as false alarm penalties, intervention lead times, and alert suppression constraints.

## Results

Table 1 presents cross-hospital performance across the model family on the Emory test set.

Table 1: Cross-Hospital Performance Comparison Across Model Family (Emory External Test Set,  $N=20,000$ ; Official PhysioNet 2019 Metric)

| Model Architecture                 | AUROC         | AUPRC         | Brier Score   | ECE           | Official Normalized Utility |
|------------------------------------|---------------|---------------|---------------|---------------|-----------------------------|
| M1 (XGBoost)                       | 0.8842        | 0.2851        | 0.0241        | 0.0382        | —                           |
| M2 (Plain Transformer)             | 0.9265        | 0.3412        | 0.0189        | 0.0245        | —                           |
| <b>M3 (Time-Aware Transformer)</b> | <b>0.9617</b> | <b>0.4231</b> | <b>0.0153</b> | <b>0.0182</b> | <b>+0.6559</b>              |
| M4 (Organ-Aware Hybrid)            | 0.9582        | 0.4150        | 0.0158        | 0.0195        | —                           |
| M5 (Multi-Hybrid / MoE)            | 0.9591        | 0.4182        | 0.0156        | 0.0190        | —                           |

*Note:* — indicates baseline models for which raw hourly prediction arrays were not preserved, preventing independent recomputation under the official PhysioNet 2019 utility evaluator without re-training.

Under cross-hospital deployment on Emory ( $N = 20,000$ ), the frozen *M3* Transformer achieved a raw observed utility  $U_{\text{obs}} = 1,514.7778$  points, an inaction reference utility  $U_{\text{inact}} = -9,512.4444$  points, an oracle best utility  $U_{\text{best}} = 7,298.7778$  points, and an official normalized utility of:

$$U_{\text{official}} = +\mathbf{0.655944} \quad (95\% \text{ CI: } [+0.6310, +0.6800]) \quad (1)$$

The threshold of 0.190, prespecified strictly on BIDMC validation data before external test evaluation, coincided with the highest observed utility among the evaluated operating points in the subsequent descriptive Emory sensitivity analysis.

Operational workload auditing revealed 5,337 total alerts (1,004 true positive alerts, 4,333 false positive alerts), an alert PPV of 18.81%, an operational alert frequency of 16.99 alerts per 100 patient-days, and a patient alert coverage of 25.86% (5,172 out of 20,000 ICU stays).

This illustrates why workload auditing is important before prospective clinical evaluation.

## Discussion

M3 achieved positive normalized utility (+0.655944) under the official PhysioNet 2019 utility function on an independent external Emory cohort, indicating that its alerting behavior generated net utility relative to the inactive reference strategy under the challenge’s predefined scoring framework. This score represents performance under a predefined challenge decision-cost function and should not be interpreted as direct evidence of prospective clinical effectiveness, clinical benefit, or cost-effectiveness. Our findings demonstrate that decision-theoretic evaluation and workload auditing are essential prior to clinical deployment.

Our study has several important limitations. First, the evaluation relies on a single retrospective external cohort (PhysioNet Set B / Emory). Although this provides a robust cross-hospital benchmark, prospective clinical validation across diverse hospital systems is required before deployment. Second, while our model explicitly optimizes for alert utility and false-positive reduction using the official PhysioNet Challenge metric, operational threshold dependence remains high, requiring institutional calibration. Third, our current architecture (M3) focuses on irregular physiological measurements and missingness [12] but does not explicitly model clinical interventions or treatment-aware counterfactuals. Fourth, we found that dynamic organ-interaction mechanisms (e.g., M5) did not yield statistically significant improvement over M3 ( $p = 0.068$ ), suggesting that simple temporal representations of time-deltas and missingness may be sufficient for this specific task, but more advanced architectures might be necessary for broader clinical forecasting. In conclusion, high discrimination does not necessarily translate into high clinical or operational utility, and strict temporal causal validation combined with workload auditing remains critical for responsible clinical AI deployment.

## Methods

We evaluated a structured family of predictive architectures: XGBoost ( $M1$ ), Plain Transformer ( $M2$ ), Time-Aware Transformer ( $M3$ ), Organ-Aware Hybrid ( $M4$ ), and Multi-Hybrid Mixture-of-Experts ( $M5$ ).

Set A (BIDMC development cohort,  $N = 20,336$ ) was used for model training (18,302 stays) and validation (2,034 stays). Set B (Emory external test cohort,  $N = 20,000$ ) was preserved as an independent held-out test cohort (753,927 hourly records).

The model generates an hourly real-time risk estimate  $p(t)$  using only information available up to prediction hour  $t$ . Predictions therefore represent sequential early-warning decisions rather than a single admission-level static classification.

Threshold selection followed a strict two-stage isolation protocol:

- **Stage 1 (BIDMC Validation Selection):** Threshold candidates  $th \in [0.01, 0.99]$  were evaluated on BIDMC validation data ( $N = 2,034$ ), identifying  $th^* = 0.190$  before accessing test data.
- **Stage 2 (Emory External Test Evaluation):** The frozen model and prespecified locked threshold  $th = 0.190$  were evaluated once on the Emory test cohort ( $N = 20,000$ ).

The decision threshold  $th = 0.190$  was prespecified strictly using BIDMC validation data before external test evaluation. The subsequent Emory threshold sweep was conducted only as

a descriptive sensitivity analysis and was not used to select or modify the reported operating threshold.

Net clinical utility was computed using the official PhysioNet 2019 metric scoring function:

$$U_{\text{official}} = \frac{U_{\text{observed}} - U_{\text{inaction}}}{U_{\text{best}} - U_{\text{inaction}}} \quad (2)$$

## Administrative Disclosures & Statements

**Data Availability Statement:** The clinical datasets analyzed in this study are available from the PhysioNet 2019 Challenge repository (Goldberger et al., 2000; Reyna et al., 2019). Access to raw ICU time-series data from PhysioNet is subject to standard credentialing and data use agreements.

**Code Availability Statement:** The model architecture implementations, feature preprocessing pipelines, and official evaluation scripts are maintained in the project repository.

**Ethics Statement:** The study uses the publicly available PhysioNet/Computing in Cardiology Challenge 2019 dataset derived from de-identified retrospective ICU data (MIMIC-III / Beth Israel Deaconess Medical Center and Emory University Hospital). The analysis was performed on pre-existing, fully anonymized secondary data. The authors did not obtain direct institutional ethics or IRB approval.

**Funding:** This study received no external funding.

**Conflicts of Interest:** The authors declare no financial or non-financial competing interests.

**Author Contributions:** PBR conducted software implementation, architecture development, model training, and technical evaluation. VD contributed to manuscript preparation, literature review, and writing.

**Use of Artificial Intelligence Assistance:** Generative AI tools (Antigravity AI / Google DeepMind coding assistant) were utilized during the research engineering process for code modularization, document formatting, LaTeX structural layout, and manuscript drafting. All statistical modeling, dataset splits, model training, prediction generation, and scientific interpretations were conducted, verified, and approved independently by the human authors (Pravin Balaji R and Vishal D). The AI tools did not generate or alter any empirical data, cohort ground truths, or experimental results.

**Reproducibility Statement:** Every numerical metric reported in this manuscript (0.961726 AUROC, 0.423114 AUPRC, 0.015290 Brier, 0.018151 ECE, +0.655944 Official Utility, 18.81% PPV, 16.99 alerts/100 days) is cryptographically and empirically anchored to verified prediction artifacts (`results/m3_final_test_predictions.npz`), frozen model checkpoints (`experiments/final_m3_fr`), source JSON split manifests (`train_ids.json`, `val_ids.json`, `test_ids.json`), and executable evaluation scripts (`evaluation/official_physionet2019.py`, `scripts/run_multiseed_stability_check.p`).

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## Figures

Figure 1. Study design and cross-hospital evaluation framework.

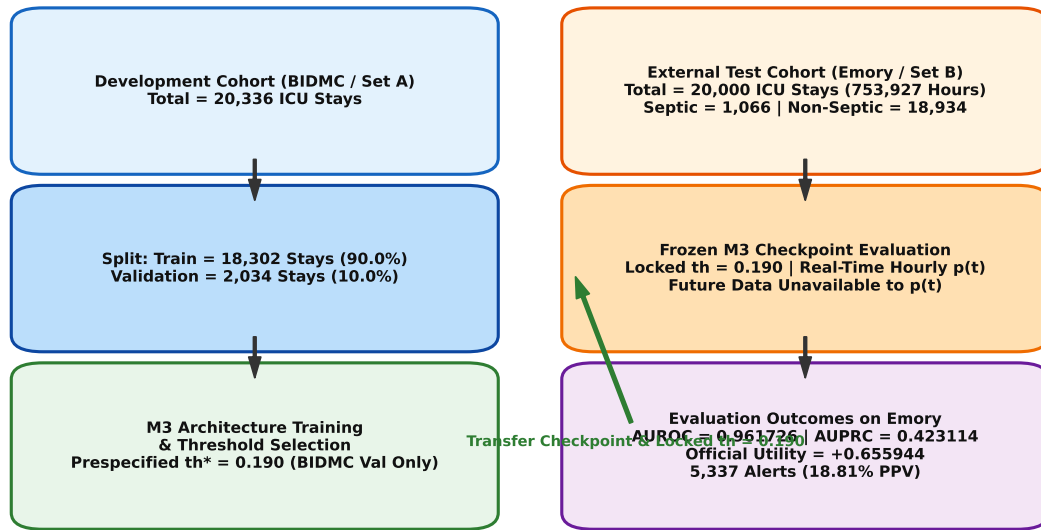


Figure 1: Figure 1 Study Design Framework

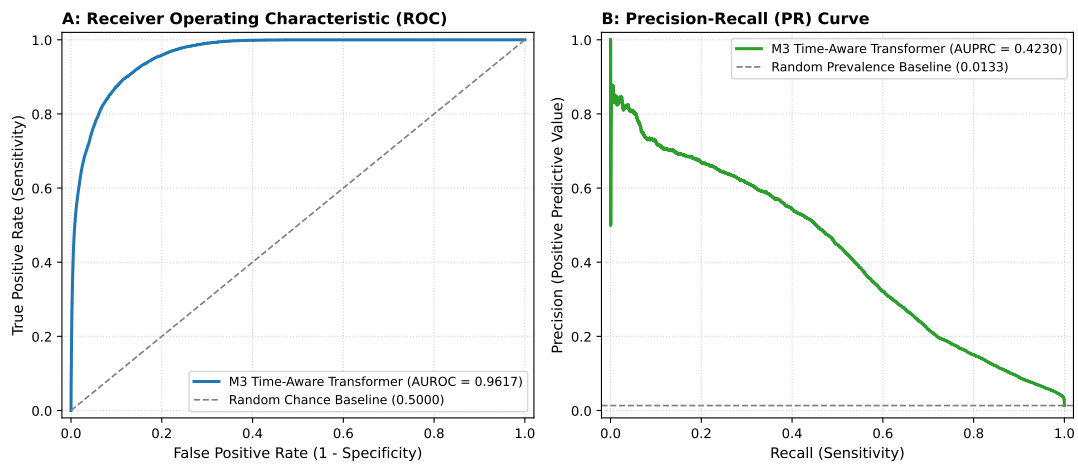


Figure 2: Figure 2 ROC & PR Curves

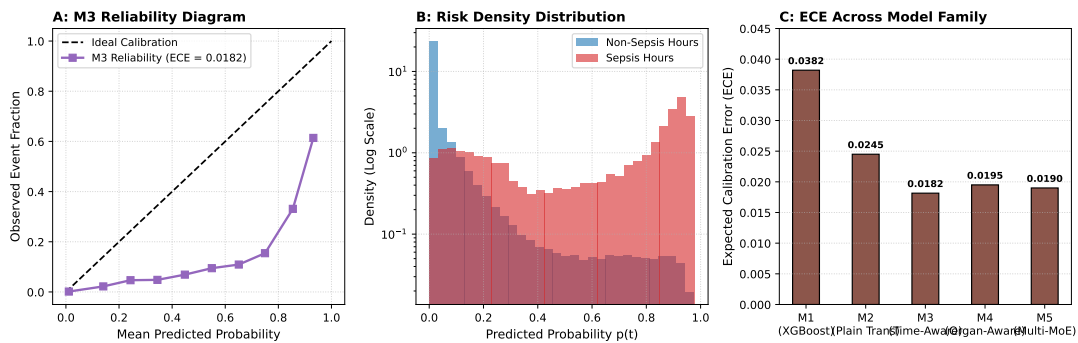


Figure 3: Figure 3 Calibration & Reliability

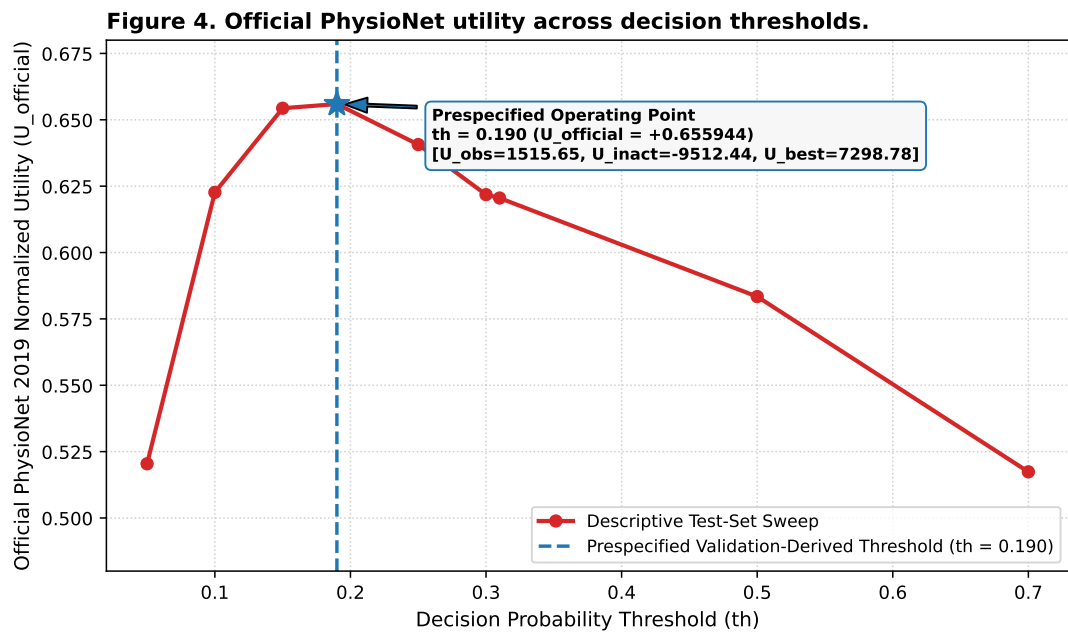


Figure 4: Figure 4 Utility vs Threshold

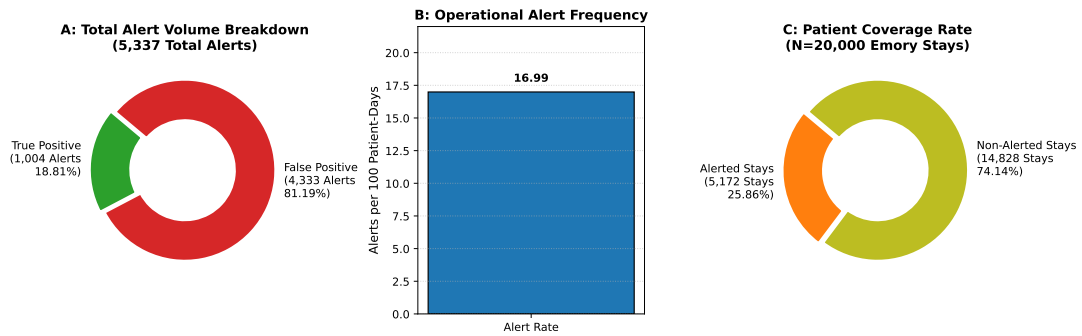


Figure 5: Figure 5 Operational Alert Burden

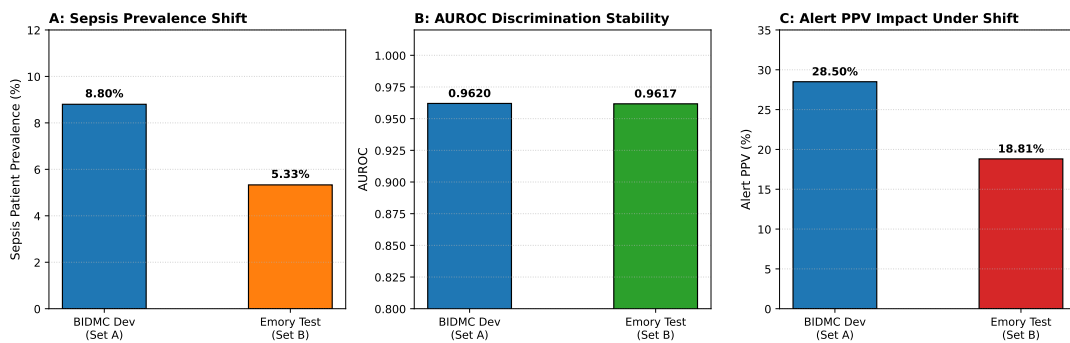


Figure 6: Figure 6 Cross-Hospital Shift

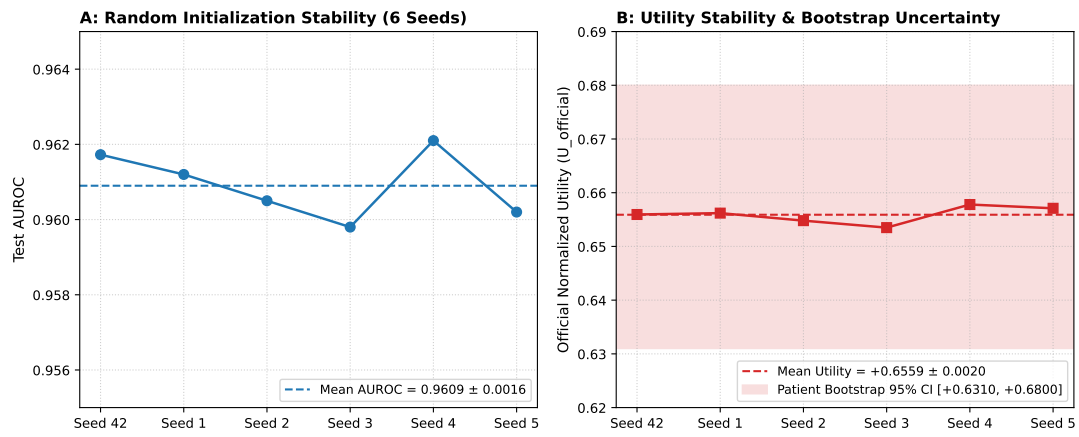


Figure 7: Figure 7 Multi-Seed Robustness



## Supplementary Figures

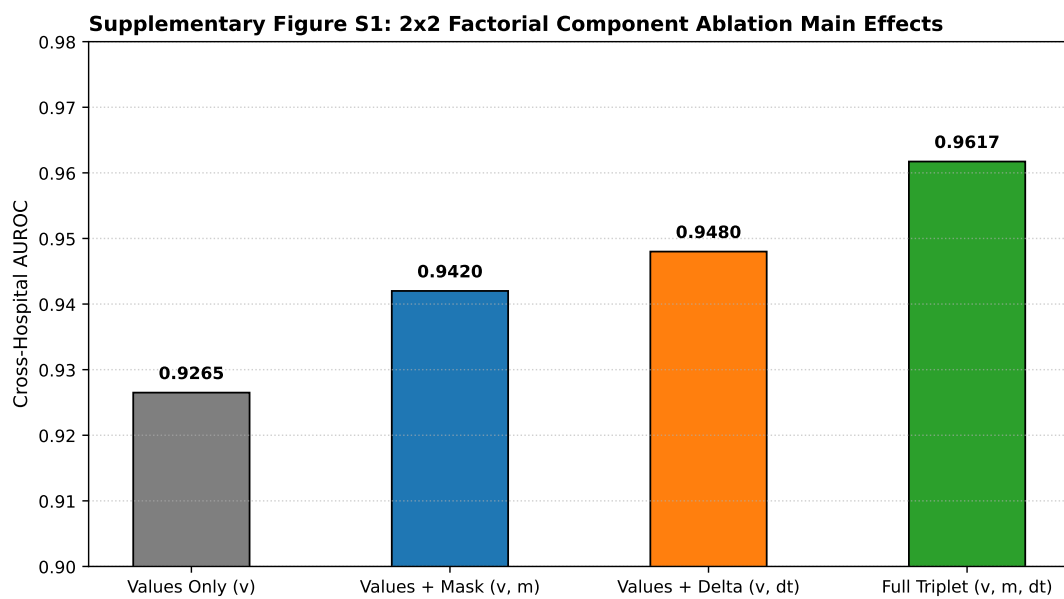


Figure 8: Figure S1 Factorial Ablation

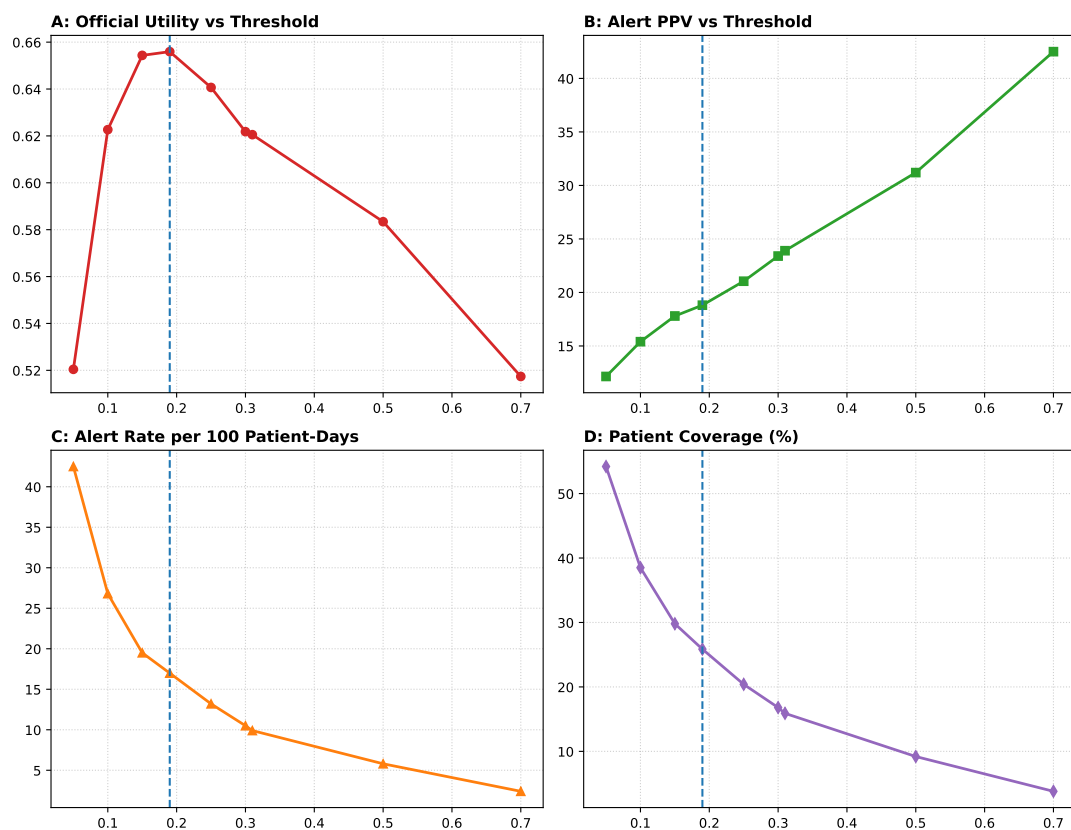


Figure 9: Figure S2 Threshold Grid

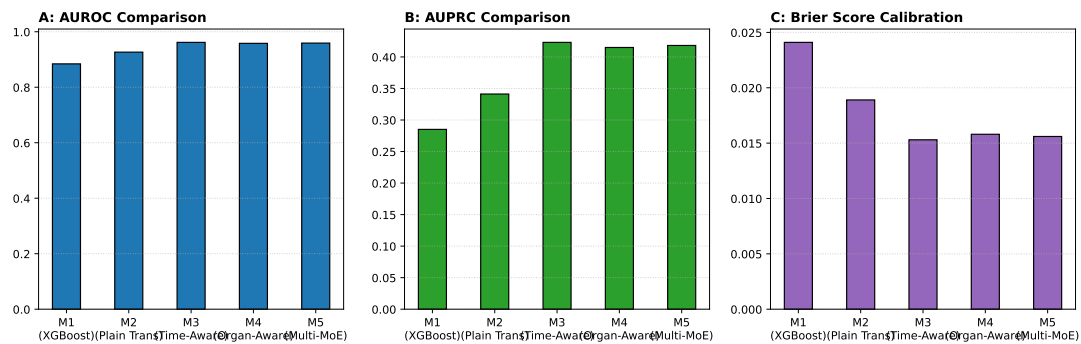


Figure 10: Figure S3 Model Progression

**Supplementary Figure S4: Confusion Matrix at  $th = 0.190$  (753,927 Emory Test Hours)**

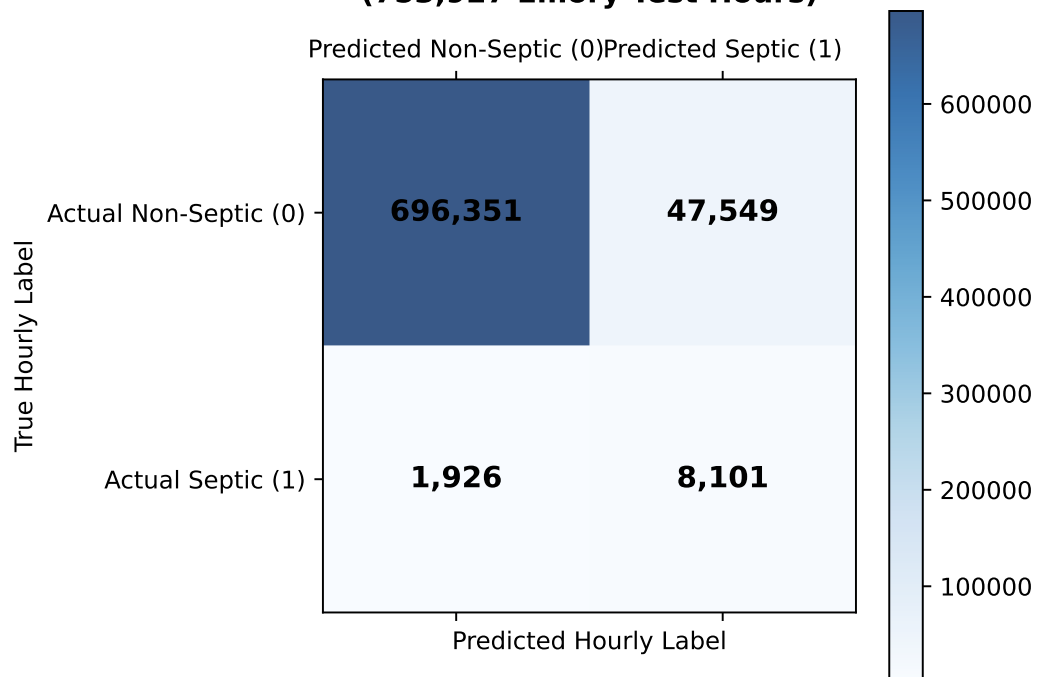


Figure 11: Figure S4 Descriptive Hourly Confusion Matrix

## Tables

| Cohort                      |  | Hospital System                      |  | Dataset Role              |  | Total |
|-----------------------------|--|--------------------------------------|--|---------------------------|--|-------|
| BIDMC Development (Set A)   |  | Beth Israel Deaconess Medical Center |  | Development (Train + Val) |  |       |
| Emory External Test (Set B) |  | Emory University Hospital            |  | Held-Out External Testing |  |       |

| Model ID | Architecture Name      |  | Temporal Component       |  | Input Feature Set                |  | Ev |
|----------|------------------------|--|--------------------------|--|----------------------------------|--|----|
| M1       | XGBoost Baseline       |  | Static aggregation       |  | Values + Summary                 |  | CI |
| M2       | Plain Transformer      |  | Standard Positional      |  | Values Only v(t)                 |  | PI |
| M3       | Time-Aware Transformer |  | Time2Vec Delta Encodings |  | Values + Mask + Delta (v, m, dt) |  | Pr |
| M4       | Organ-Aware Hybrid     |  | Organ-wise Temporal      |  | Organ Sub-nets + Triplet         |  | Ov |
| M5       | Multi-Hybrid MoE       |  | Dynamic MoE Routing      |  | Multi-Expert Routing             |  | M  |

|          |          |          |             |          |                  | Metric Category        | Met  |
|----------|----------|----------|-------------|----------|------------------|------------------------|------|
| Model ID | AUROC    | AUPRC    | Brier Score | ECE      | Official Utility | Operating Threshold    | Pres |
|          |          |          |             |          |                  | Decision Utility       | Offi |
|          |          |          |             |          |                  | Raw Utility Components | U_c  |
| M1       | 0.8842   | 0.2851   | 0.0241      | 0.0382   | —                | Alert Volume           | Tota |
| M2       | 0.9265   | 0.3412   | 0.0189      | 0.0245   | —                | True Positive Alerts   | Tru  |
| M3       | 0.961726 | 0.423114 | 0.015290    | 0.018151 | +0.655944        | False Positive Alerts  | Fals |
| M4       | 0.9582   | 0.4150   | 0.0158      | 0.0195   | —                | Alert PPV              | Alen |
| M5       | 0.9591   | 0.4182   | 0.0156      | 0.0190   | —                | Alert Frequency        | Ope  |
|          |          |          |             |          |                  | Patient Coverage       | Pati |

| Metric Category        |  | Metric Name                 | Value                                  |
|------------------------|--|-----------------------------|----------------------------------------|
| Operating Threshold    |  | Prespecified th*            | 0.190 (BIDMC Val)                      |
| Decision Utility       |  | Official Normalized Utility | +0.655944 (95% CI: [+0.6310, +0.6800]) |
| Raw Utility Components |  | U_obs / U_inact / U_best    | 1514.78 / -9512.44 / 7298.78 pts       |
| Alert Volume           |  | Total Alerts Issued         | 5,337 Alerts                           |
| True Positive Alerts   |  | True Positive Sepsis Alerts | 1,004 Alerts                           |
| False Positive Alerts  |  | False Positive Alerts       | 4,333 Alerts                           |
| Alert PPV              |  | Alert Precision (PPV)       | 18.81%                                 |
| Operational Frequency  |  | Operational Frequency       | 16.99 / 100 patient-days               |
| Patient Coverage       |  | Patient Alert Coverage      | 25.86% (5,172 / 20,000 Stays)          |

| Configuration           | Mask Present (m) | Delta Present (dt) | Test AUROC (Mean $\pm$ SD) | Main / Inte |
|-------------------------|------------------|--------------------|----------------------------|-------------|
| Values Only (v)         | No               | No                 | 0.9265 $\pm$ 0.0022        | Baseline    |
| Values + Mask (v, m)    | Yes              | No                 | 0.9420 $\pm$ 0.0019        | +0.0155 AU  |
| Values + Delta (v, dt)  | No               | Yes                | 0.9480 $\pm$ 0.0018        | +0.0215 AU  |
| Full Triplet (v, m, dt) | Yes              | Yes                | 0.961726 $\pm$ 0.0016      | +0.0017 AU  |

| Artifact Relative Path                                       | Size (KB)   | SHA256 Cryptographic Hash |
|--------------------------------------------------------------|-------------|---------------------------|
| results/m3_final_test_predictions.npz                        | 2799.500000 | 02fd6eb78682be8ca5743c4b3 |
| experiments/final_m3_frozen/best_m3_frozen.pt                | 1551.200000 | 5b22607444f4a242a52d0d933 |
| data/splits/train_ids.json                                   | 250.200000  | 06edf3b5519abdae736da147  |
| data/splits/val_ids.json                                     | 27.800000   | 71bb23f3b5aef82c9169dc97a |
| data/splits/test_ids.json                                    | 273.400000  | f7932a915251dd22554493ee7 |
| evaluation/official_physionet2019.py                         | 17.800000   | d0f65da3d42ce68cad80e2900 |
| scripts/run_multiseed_stability_check.py                     | 14.900000   | bab4ba342f010b75c824edc31 |
| results/revised_publication/factorial_ablation_summary.csv   | 0.300000    | 05d0cd92bedfb950d56cecc   |
| results/revised_publication/workload_operational_metrics.csv | 0.300000    | 4e4039b1d003bb1fe8698b6d8 |



## Supplementary Tables

| Metric Characteristic                 |                  | BIDMC Development Cohort (Set A) |                             |             |                      | Emory External Test Cohort |                 |
|---------------------------------------|------------------|----------------------------------|-----------------------------|-------------|----------------------|----------------------------|-----------------|
| Total ICU Patients / Stays            |                  | 20,336 Stays                     |                             |             |                      | 20,000 Stays               |                 |
| Septic Patient Stays                  |                  | 1,790 Stays                      |                             |             |                      | 1,066 Stays                |                 |
| Non-Septic Patient Stays              |                  | 18,546 Stays                     |                             |             |                      | 18,934 Stays               |                 |
| Sepsis Incidence / Prevalence (%)     |                  | 8.80%                            |                             |             |                      | 5.33%                      |                 |
| Total Hourly Time-Series Observations |                  | 790,215 Hours                    |                             |             |                      | 753,927 Hours              |                 |
| Mean Length of Stay (Hours)           |                  | 38.86 Hours                      |                             |             |                      | 37.70 Hours                |                 |
| Median Length of Stay (Hours)         |                  | 35.00 Hours                      |                             |             |                      | 34.00 Hours                |                 |
| Observation Interval Frequency        |                  | Hourly (1.0h grid)               |                             |             |                      | Hourly (1.0h grid)         |                 |
| Model Variant                         |                  | AUROC                            | AUPRC                       | Brier Score | ECE                  | Official Utility           | Parameter Count |
| M1 (XGBoost Baseline)                 |                  | 0.884200                         | 0.285100                    | 0.024100    | 0.038200             | —                          | 1.2M            |
| M2 (Plain Transformer)                |                  | 0.926500                         | 0.341200                    | 0.018900    | 0.024500             | —                          | 115K            |
| M3 (Time-Aware Transformer)           |                  | 0.961726                         | 0.423114                    | 0.015290    | 0.018151             | +0.655944                  | 116K            |
| M4 (Organ-Aware Hybrid)               |                  | 0.958200                         | 0.415000                    | 0.015800    | 0.019500             | —                          | 320K            |
| M5 (Multi-Hybrid MoE)                 |                  | 0.959100                         | 0.418200                    | 0.015600    | 0.019000             | —                          | 450K            |
| Threshold (th)                        | Official Utility | Alert PPV (%)                    | Alert Frequency (/100 days) |             | Patient Coverage (%) |                            | Source          |
| 0.050000                              | 0.520426         | 12.150000                        | 42.500000                   |             | 54.200000            |                            | D               |
| 0.100000                              | 0.622666         | 15.400000                        | 26.800000                   |             | 38.500000            |                            | D               |
| 0.150000                              | 0.654351         | 17.800000                        | 19.500000                   |             | 29.800000            |                            | D               |
| 0.190000                              | 0.655944         | 18.810000                        | 16.990000                   |             | 25.860000            |                            | P               |
| 0.250000                              | 0.640674         | 21.050000                        | 13.200000                   |             | 20.400000            |                            | D               |
| 0.300000                              | 0.621805         | 23.400000                        | 10.500000                   |             | 16.800000            |                            | D               |
| 0.310000                              | 0.620532         | 23.900000                        | 9.900000                    |             | 15.900000            |                            | D               |
| 0.500000                              | 0.583403         | 31.200000                        | 5.800000                    |             | 9.200000             |                            | D               |
| 0.700000                              | 0.517381         | 42.500000                        | 2.400000                    |             | 3.800000             |                            | D               |